

# Midpoint preservation on a symmetric lens

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

Turns bounded reflection symmetry of equal-radius lenses into midpoint preservation.

## Statement

Assume the midpoint of  $x,y$  belongs to  $\text{symmetricLens}(x,y,r)$ , and likewise the midpoint of  $x',y'$  belongs to  $\text{symmetricLens}(x',y',r)$ , as expressed by the two half-distance inequalities. Any isometry equivalence between these lenses maps  $\text{midpoint}(x,y)$  to  $\text{midpoint}(x',y')$ .

## Assumptions

Real normed vector spaces; an isometry equivalence between symmetric lenses of the same radius  $r$ ; both half-distances between the respective foci are at most  $r$ .

## Conclusion

The isometry maps the midpoint of the source foci to the midpoint of the target foci.

## Proof idea

A symmetric lens is bounded and invariant under point reflection about the midpoint of its foci. The center-transport theorem therefore sends the source midpoint to the target midpoint.

## Proof steps

1. Use the half-distance inequalities to place both midpoints in their respective lenses.
2. Bound the source lens by one of its defining closed balls.
3. Use reflection invariance of each lens about the corresponding midpoint.
4. Apply center transport to the lens isometry and erase the subtype wrapper.

## Main citations

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.mapsTo_pointReflection_symmetricLens`

The proof or construction of Midpoint preservation on a symmetric lens uses the project declaration “A symmetric lens is invariant under reflection about its midpoint” at the indicated step.

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

The proof or construction of Midpoint preservation on a symmetric lens uses the project declaration “Center transport under bounded point-reflection symmetry” at the indicated step.

`MathlibAnnex.symmetricLens`

The proof or construction of Midpoint preservation on a symmetric lens uses the project declaration “Symmetric lens” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.midpoint_mem_symmetricLens`

The proof or construction of Midpoint preservation on a symmetric lens uses the project declaration “The midpoint belongs to a sufficiently large symmetric lens” at the indicated step.

## Lean source signature (exact)

```
theorem map_midpoint_of_symmetricLens
  {x y : E} {x' y' : F} {r : ℝ}
  (hxy : 2-1 * dist x y ≤ r) (hxy' : 2-1 * dist x' y' ≤ r)
  (f : symmetricLens x y r ≈i symmetricLens x' y' r) :
  ((f (midpoint ℝ x y, midpoint_mem_symmetricLens hxy) :
  symmetricLens x' y' r) : F) = midpoint ℝ x' y'
```

[Read exact MathlibAnnex v0.4.0 source](#)

## Exact Card identity

Stable Card ID: 4a485c9aabdc6f37ae97b47bfba6021a6bedf685894a44120f0290571e459c36

Card SHA-256: aa07f5efc7a9d59552a046091c57dfad6548ee065751704b012ec1f00954522b

Approved exposition SHA-256: d744bad64277ccb9a4d5ef0cf4fc1fb458eb1342e95462e63c0e9c74f4af8602

AI-assisted material. Selected, edited, and reviewed by Ryotaro Tanaka. The recorded review is source-exposition correspondence, not an independent proof audit.